STACKS, EXAMPLES OF DEFORMATIONS: LEMMA-FIRST-ORDER-COMPLETION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2382-2393 of sources/stacks-examples-defos/pinned-original-source.tex; proof span is lines 2395-2489, the complete author proof (93 body lines).

Editorial note. Navigation stubs: 8 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 0 of queue_examples-defos.json (pb=3523, stmt 2382-2393, proof 2395-2489).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 00S1: sources/stacks-examples-defos/pinned-original-source.tex, source lines 2489–2489. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DYC: sources/stacks-examples-defos/pinned-original-source.tex, source lines 2489–2489. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00MA: sources/stacks-examples-defos/pinned-original-source.tex, source lines 2489–2489. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00S0: sources/stacks-examples-defos/pinned-original-source.tex, source lines 2489–2489. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DYV: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1309–1318. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DYJ: sources/stacks-examples-defos/pinned-original-source.tex, source lines 2489–2489. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05EC: sources/stacks-examples-defos/pinned-original-source.tex, source lines 2489–2489. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 087Q: sources/stacks-examples-defos/pinned-original-source.tex, source lines 2489–2489. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let k be a field. Let B be a finite type k-algebra. Let $J \subset B$ be an ideal such that $\mathrm{Spec}(B) \to \mathrm{Spec}(k)$ is smooth on the complement of $V(J)$. Let N be a finite B-module. Then there is a canonical bijection*

$$Exal_k(B, N) \to Exal_k(B^\wedge, N^\wedge)$$

Here $B^\wedge$ and $N^\wedge$ are the J-adic completions.

Proof. The map is given by completion: given $0 \to N \to C \to B \to 0$ in $\mathrm{Exal}_k(B, N)$ we send it to the completion $C^\wedge$ of C with respect to the inverse image of J. Compare with the proof of Lemma 0DYV.

Since $k \to B$ is of finite presentation the complex $NL_{B/k}$ can be represented by a complex $N^{-1} \to N^0$ where N^i is a finite B-module, see Algebra, Section 00S0 and in particular Algebra, Lemma 00S1. As B is Noetherian, this means that $NL_{B/k}$ is pseudo-coherent. For $g \in J$ the k-algebra B_g is smooth and hence $(NL_{B/k})_g = NL_{B_g/k}$ is quasi-isomorphic to a finite projective B-module sitting in degree 0. Thus $\mathrm{Ext}^i_B(NL_{B/k}, N)_g = 0$ for $i \geq 1$ and any B-module N. By More on Algebra, Lemma 0DYJ we conclude that

$$\mathrm{Ext}^1_B(NL_{B/k}, N) \longrightarrow \lim_n \mathrm{Ext}^1_B(NL_{B/k}, N/J^nN)$$

is an isomorphism for any finite B-module N.

Injectivity of the map. Suppose that $0 \to N \to C \to B \to 0$ is in $\mathrm{Exal}_k(B, N)$ and maps to zero in $\mathrm{Exal}_k(B^\wedge, N^\wedge)$. Choose a splitting $C^\wedge = B^\wedge \oplus N^\wedge$. Then the induced map $C \to C^\wedge \to N^\wedge$ gives maps $C \to N/J^nN$ for all n. Hence we see that our element is in the kernel of the maps

$$\mathrm{Ext}^1_B(NL_{B/k}, N) \to \mathrm{Ext}^1_B(NL_{B/k}, N/J^nN)$$

for all n. By the previous paragraph we conclude that our element is zero.

Surjectivity of the map. Let $0 \to N^\wedge \to C' \to B^\wedge \to 0$ be an element of $\mathrm{Exal}_k(B^\wedge, N^\wedge)$. Pulling back by $B \to B^\wedge$ we get an element $0 \to N^\wedge \to C'' \to B \to 0$ in $\mathrm{Exal}_k(B, N^\wedge)$. we have

$$\mathrm{Ext}^1_B(NL_{B/k}, N^\wedge) = \mathrm{Ext}^1_B(NL_{B/k}, N) \otimes_B B^\wedge = \mathrm{Ext}^1_B(NL_{B/k}, N)$$

The first equality as $N^\wedge = N \otimes_B B^\wedge$ (Algebra, Lemma 00MA) and More on Algebra, Lemma 087Q. The second equality because $\mathrm{Ext}^1_B(NL_{B/k}, N)$ is J-power torsion (see above), $B \to B^\wedge$ is flat and induces an isomorphism $B/J \to B^\wedge/JB^\wedge$, and More on Algebra, Lemma 05EC. Thus we can find a $C \in \mathrm{Exal}_k(B, N)$ mapping to C'' in $\mathrm{Exal}_k(B, N^\wedge)$. Thus

$$0 \to N^\wedge \to C' \to B^\wedge \to 0 \quad \text{and} \quad 0 \to N^\wedge \to C^\wedge \to B^\wedge \to 0$$

are two elements of $\mathrm{Exal}_k(B^\wedge, N^\wedge)$ mapping to the same element of $\mathrm{Exal}_k(B, N^\wedge)$. Taking the difference we get an element $0 \to N^\wedge \to C' \to B^\wedge \to 0$ of $\mathrm{Exal}_k(B^\wedge, N^\wedge)$ whose image in $\mathrm{Exal}_k(B, N^\wedge)$ is zero. This means there exists

$$\begin{array}{ccccccccc} 0 & \longrightarrow & N^\wedge & \longrightarrow & C' & \longrightarrow & B^\wedge & \longrightarrow & 0 \\ & & & & \uparrow{\scriptstyle \sigma} & \nearrow & & & \\ & & & & B & & & & \end{array}$$

Let $J' \subset C'$ be the inverse image of $JB^\wedge \subset B^\wedge$. To finish the proof it suffices to note that σ is continuous for the J-adic topology on B and the J'-adic topology on C' and that C' is J'-adically complete by Algebra, Lemma 0DYC (here we also use that C' is Noetherian; small detail omitted). Namely, this means that σ factors through the completion $B^\wedge$ and $C' = 0$ in $\mathrm{Exal}_k(B^\wedge, N^\wedge)$. $\square$